LOLA: LLM-Assisted Online Learning Algorithm for Content Experiments

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Replicated and Presented by Xizhi Liu and Shaoqiang Liu | DOTE6635, 2026

Digital content consumption is growing at an unprecedented pace, creating both opportunities and challenges for media firms. In this hyper-competitive landscape, publishers must continuously identify the most appealing articles, craft the most attractive headlines, and design the most engaging packaging to capture audience attention. The key metric driving these decisions is **Click-Through Rate (CTR)** — a critical measure of how effectively headlines convert impressions into clicks.

This research explores whether Large Language Models (LLMs) can revolutionize how media companies conduct content experimentation, potentially reducing costs while maintaining or improving the quality of headline selection.

Traditional Approaches to Content Testing

A/B Testing: Explore-and-Commit

The traditional A/B testing framework uses fixed traffic allocation across treatment arms. After an exploration phase, the publisher commits to the best-performing arm. This approach is widely used throughout the publishing industry, exemplified by platforms like the New York Times ABRA system and Upworthy's headline testing methodology.

Advantages

- Statistically unbiased estimates
- Reliable inference with sufficient traffic volume
- Simple interpretation for stakeholders
- Well-understood statistical properties

Limitations

- Equal traffic allocation to poor-performing arms
- High regret during exploration phase
- Particularly problematic for short-lived content like news articles
- Requires large sample sizes for statistical power

Online Learning: Bandit Algorithms

Modern online learning approaches use multi-armed bandit algorithms that dynamically allocate traffic based on past performance, content features, and uncertainty quantification. Major platforms like Yahoo News and the New York Times have implemented contextual bandit frameworks.

Advantages

- Lower regret compared to traditional A/B tests
- Gradually reduce exploration as performance becomes clear
- Faster convergence to best-performing arms
- Continuous optimization without discrete phases

Limitations

- Cold-start problem: arms initialized as equally good
- Early traffic potentially wasted on inferior options
- Risk of converging to suboptimal arms
- Designed for large-traffic regimes, not limited horizons

Core Research Questions

1 Can LLMs directly predict which headline is most appealing?

Is it possible to leverage the semantic understanding capabilities of large language models to predict human preferences without extensive experimentation?

2 Can LLMs reduce experimentation costs?

Can LLM-assisted methods achieve comparable or better performance than traditional approaches while requiring fewer impressions and less traffic allocation to inferior options?

3 Can LLMs replace traditional experiments entirely?

Is there a scenario where LLM-based prediction becomes sufficiently accurate that it obviates the need for A/B testing or bandit algorithms altogether?

Dataset: Upworthy Headline Experiments (2013–2015)

The researchers analyzed headline testing data from Upworthy, a digital media company known for its rigorous experimentation practices. The original dataset contained 32,487 A/B tests across 150,817 packages, with 538 million impressions generating 8.18 million clicks. After filtering to headline-specific tests, the analysis focused on **17,681 tests** covering **77,245 headlines** with **277 million impressions** and **3.74 million clicks**. The average CTR across all tests was 1.52%, with an average of approximately 4.37 headlines per test.

The prediction task was straightforward: for each A/B test, identify the headline with the highest observed CTR. The data was split into 70% training, 10% for hyperparameter tuning, and 20% for testing to ensure robust evaluation of model performance. Specifically, in this section, there were 12,376 headline tests as a training set and 3,263 headline tests as a test set.

17,681

A/B Tests

Headline-specific experiments

77,245

Headlines

Unique headline variants

277M

Impressions

Total display occurrences

3.74M

Clicks

Observed user actions

Pure LLM-Based Methods Evaluated

The researchers systematically evaluated three distinct LLM-based approaches.
We replicated and extended the original evaluation using the latest GPT models and reported partial results.
See the GitHub repository for implementation details and additional findings: https://github.com/GHCharlesLau/AI-for-research-nexusAI

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Prompt-Based Approaches

Direct interaction with LLMs through carefully crafted prompts, testing both zero-shot and in-context learning variants.

- Zero-shot prompting: LLM directly asked to identify catchiest headline
- In-context learning: Demonstrations provided before query
- Role prompting: LLM assigned specific expert persona
- Temperature and formatting controls

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Embedding-Based CTR Prediction

Using LLM-generated embeddings as input features for supervised machine learning models.

- OpenAI embedding models (256-dim and 3072-dim)
- Linear regression with embedding features
- Multilayer perceptron (MLP) architectures
- End-to-end CTR prediction framework

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Fine-Tuned LLM

Custom training of LLM on headline prediction task (covered in detail by next presenter).

- Full model fine-tuning on labeled dataset
- Task-specific architecture modifications
- Hyperparameter optimization
- Comprehensive performance evaluation

Prompt-Based Method 1: Zero-Shot Prompting

The first prompt-based approach tested was **zero-shot prompting**, where the LLM is directly asked to identify the catchiest headline without any prior examples or demonstrations. The prompt structure was straightforward: "Which headline is the catchiest?" The researchers also experimented with role prompting, where the LLM was assigned the persona of a marketing expert to see if this improved performance.

Experimental Controls

- Temperature set to 0 for deterministic outputs
- Randomized order of headline options across tests
- Different marker types (letters, numbers, symbols)
- Multiple model versions tested

Models Tested

We replicate the original evaluation on:

- GPT-3.5
- GPT-4
- GPT-4o

and extend the analysis to include:

- GPT-5.2

System

You are an expert in digital media marketing and content strategy, specializing in optimizing headlines for maximum click-through rates (CTR) and increasing ad revenue. Your deep understanding of reader behavior, SEO, and digital engagement allows you to evaluate headlines for their potential to capture attention and drive clicks.

I have a news article, and I need to choose the catchiest headline from the following list. By “catchiest,” I mean the headline that is most likely to generate the highest CTR, thereby increasing ad revenue for the news company. The selected headline should:

1. Capture readers’ attention immediately.
2. Appeal to emotions, curiosity, or urgency.
3. Be engaging enough to make readers want to click on the article.
4. Follow best practices for digital news, considering SEO, shareability, and intrigue.

Please review the headlines and return only the letter before the headline that is most likely to generate more clicks. **No explanation is needed. No need to return the headline, only the letter.**

User

Here are the headlines I need you to evaluate:

A. Hey Dude. If You Have An Older Brother, There's A Bigger Chance You're Gay.
B. Here's The Science, Here's The Gay. Open Your Brain, They Were Born That Way!

C. I've Got Some News For You. Being Gay Is Genetic. Being Irrationally Afraid Of Gay, Not So Much.
D. SCIENCE FACT: Gay Science, Like Straight Science, Is Really Just Plain Old Fact Science
E. If You Know Anyone Who Is Afraid Of Gay People, Here's A Cartoon That Will Ease Them Back To Reality

Assistant

E

Prompt-Based Method 2: In-Context Learning

The second prompt-based approach tested was **in-context learning**, where the LLM is provided with demonstrations before the actual query. This approach tests whether showing examples of headline sets and their correct winners helps the model learn the underlying task structure and improve prediction accuracy.

Demonstrations

Example headline sets with labeled winners provided as context

Learning

Model learns format and task structure from examples

Prediction

Applies learned patterns to new headline sets

Experimental Variations

- **Number of demonstrations:** 2 examples vs. 5 examples
- **Label correctness:** Correct labels vs. intentionally incorrect labels
- **Example selection:** Random vs. curated demonstrations
- **Formatting:** Consistent structure across all demonstrations

Models Tested

- GPT-3.5
- GPT-4
- GPT-4o
- GPT-5.2

Demonstrations

Here are examples of headlines that have worked well before:
Example 1:
A. New York's Last Chance To Preserve Its Water Supply
B. How YOU Can Help New York Stay Un-Fracked In Under 5 Minutes
C. Why Yoko Ono Is The Only Thing Standing Between New York And Catastrophic Gas Fracking

Best-performing headline: C

Example 2:
[Second demonstration]

System

You are an expert in digital media marketing and content strategy, specializing in optimizing headlines for maximum click-through rates (CTR) and increasing ad revenue. Your deep understanding of reader behavior, SEO, and digital engagement allows you to evaluate headlines for their potential to capture attention and drive clicks.

I have a news article, and I need to choose the catchiest headline from the following list. By "catchiest," I mean the headline that is most likely to generate the highest CTR, thereby increasing ad revenue for the news company. The selected headline should:

1. Capture readers' attention immediately.
2. Appeal to emotions, curiosity, or urgency.
3. Be engaging enough to make readers want to click on the article.
4. Follow best practices for digital news, considering SEO, shareability, and intrigue.

I will also provide examples of multiple headline sets that have performed well in the past, along with the best-performing headline index for each set to guide your selection.

Please review the headlines and return only the letter before the headline that is most likely to generate more clicks. **No explanation is needed. No need to return the headline, only the letter.**

User

Here are the headlines I need you to evaluate:

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Prompt-Based Results

Results for Prompt-Based Approaches with Different GPT models, the Number of Demonstrations, and Whether We Intentionally Provide Incorrect Solutions in the Demonstration

Model	n_{demo}	Incorrect label	Accuracy (%)
Random guess	—	—	33.02
GPT-3.5	0	—	29.76
GPT-3.5	2	0	34.23
GPT-3.5	2	1	31.60
GPT-3.5	5	0	30.95
GPT-3.5	5	1	30.59
GPT-4	0	—	37.85
GPT-4	2	0	39.96
GPT-4	2	1	38.77
GPT-4	5	0	39.17
GPT-4	5	1	38.28
GPT-5.2	0	—	40.82
GPT-5.2	2	0	41.62
GPT-5.2	2	1	40.73

Main Takeaways

- Performance improves with model scale (GPT-3.5 → GPT-4 → GPT-5.2), but remains capped at around 42%.
- GPT-3.5 performs at or below random guess, whereas GPT-4 and GPT-5.2 consistently outperform it.
- In-context learning slightly improves performance over zero-shot prompting.
- Increasing demonstrations from 2 to 5 does not meaningfully improve accuracy.
- Providing incorrect labels in demonstrations has little effect on performance.
- Even the best-performing configuration remains only moderately above random guess.

Embedding-Based CTR Prediction Framework

The third approach represents a fundamentally different paradigm: rather than asking the LLM to directly predict the best headline, this method uses LLMs as **feature extractors** to create rich semantic embeddings, which are then fed into traditional supervised learning models for CTR prediction.

Step 1: Embedding Generation

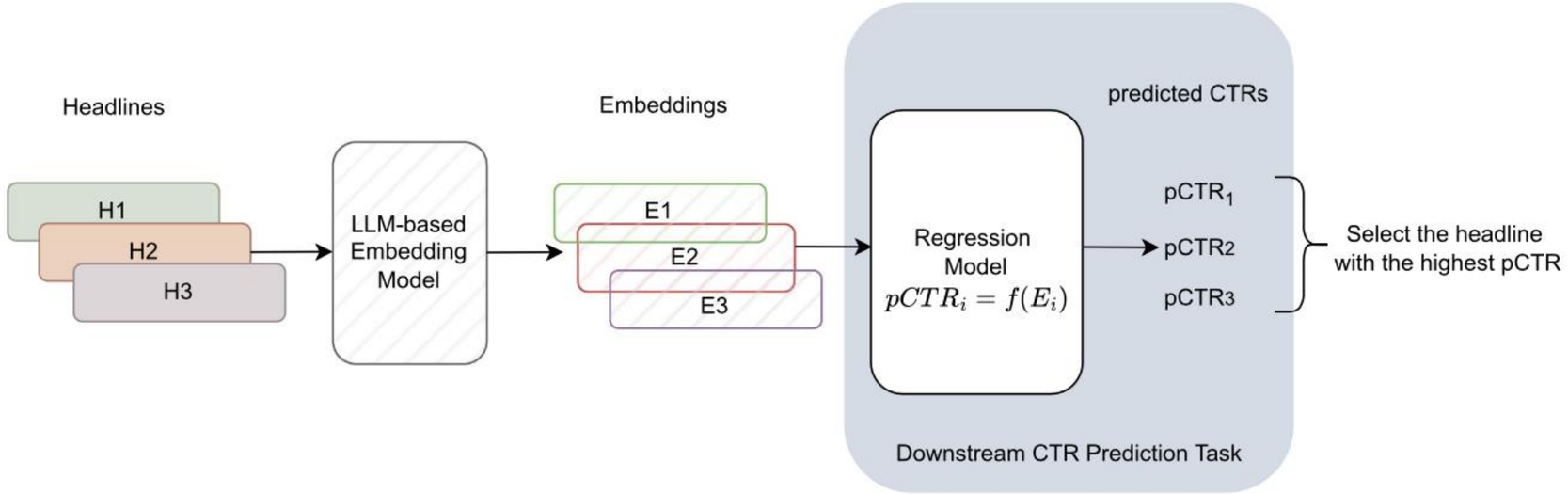
Convert each headline into a dense vector representation using OpenAI's embedding model. The embedding captures semantic meaning, tone, sentiment, and other linguistic features in a fixed-dimensional space.

Step 2: Model Training

Use the generated embeddings as input features for supervised machine learning models. The researchers tested both linear regression and multilayer perceptron (MLP) architectures, training on historical headline-test pairs with known CTR outcomes.

Step 3: Prediction & Selection

For each new A/B test, generate embeddings for all headline options, predict CTR for each using the trained model, and select the headline with the highest predicted CTR.



Embedding-Based Results

The following table summarizes performance across different embedding dimensionality and model architectures:

Embedding Model	Architecture	Accuracy
256-dimensional	Linear regression	**~46.28%** **(46.12%)**
256-dimensional	Multilayer perceptron (MLP)	~44.00%
3072-dimensional	Linear regression	~43.00%
3072-dimensional	Multilayer perceptron (MLP)	**~45.00%**

Key Observations

Supervised Learning Advantage

Embedding-based models achieved 43-46% accuracy, substantially outperforming prompting methods (37-40%). This demonstrates the value of learning from large labeled datasets rather than relying solely on LLM general knowledge.

Linear Models Compete Well

Simple linear regression performed competitively with more complex MLP architectures, suggesting that the embedding space captures features that are largely linearly separable for this task.

Dimensionality Not Critical

Higher-dimensional embeddings (3072-dim) did not consistently outperform lower-dimensional ones (256-dim), indicating that the additional dimensions may contain noise or irrelevant information for headline prediction.

Architecture Trade-offs

MLPs showed slightly better performance with 3072-dim embeddings but worse with 256-dim, suggesting complex models require sufficient feature dimensionality to avoid overfitting on limited signal.

Interpretation of Embedding Results

Why Embeddings Outperformed Prompting

- Large Labeled Dataset

  Supervised learning leveraged 17,681 labeled tests, enabling the model to learn patterns specific to headline appeal rather than relying on general knowledge

- Task-Specific Decision Rules

  The model learned which embedding dimensions and combinations predict high CTR in this specific domain

- Robust Generalization

  Training on diverse headline sets created models that generalize better to new, unseen headline variations

- No Few-Shot Limitation

  Unlike in-context learning, supervised models aren't constrained by the number of demonstrations that fit in the prompt

Despite the improvements, **accuracy remains below 50%**, meaning models still fail to identify the best headline more than half the time. This indicates that LLMs alone cannot fully replace experimentation in high-stakes content decisions.

Implications for Practice

Cost-Effectiveness

Embedding-based models could reduce experimentation costs by pre-screening poor headline options, though complete replacement remains impractical

Hybrid Approaches

Future work should explore LOLA (LLM-Assisted Online Learning Algorithm) frameworks that combine LLM predictions with bandit algorithms for adaptive traffic allocation

Refined Features

Performance might improve with domain-specific embeddings trained on headline data rather than generic OpenAI embeddings

Future Research Directions

- Testing fine-tuned LLM architectures on this dataset
- Developing LOLA frameworks that integrate LLM predictions with online learning
- Exploring cold-start solutions for new content categories
- Validating findings on news headline datasets with shorter time horizons
- Investigating multi-modal approaches incorporating article content and images